

SHAHAZAD ABDULLA

Kottayam, Kerala, India

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Summary

Results-driven Robotics and Automation Engineer with extensive R&D experience at **CSIR-NAL** and **Intel**. Proven expertise in designing and implementing robust autonomous systems, specializing in **safety-critical, end-to-end systems integration**, from low-level IoT data pipelines to high-level perception including VSLAM and sensor fusion. Recognized as a **Top 3 global contributor** to a major ROS 2 release and a key developer on the **Team Equinox (BAJA SAEINDIA)** autonomous vehicle team. Seeking a challenging internship to apply advanced skills in full-stack robotics systems integration.

Education

Saintgits College of Engineering (Autonomous)
Bachelor of Technology in Robotics and Automation Engineering

Kottayam, Kerala, India
Expected July 2027

Skills

- **Programming:** Python, C++, C, MATLAB, Bash
- **Robotics Software:** ROS 2, Gazebo, Isaac Sim, Navigation2, SLAM (RTAB-Map, VSLAM), Sensor Fusion (EKF), Digital Twins (PyBullet), micro-ROS, Control Systems (PID), URDF/SDF/XACRO
- **Hardware & Comms:** Hardware: NVIDIA Jetson, Microcontrollers (Arduino, ESP32), Motor Controllers, DACs, IMUs, RADAR
Comms & Protocols: CAN bus, MQTT, UDP, I2C, Serial
- **Computer Vision & ML:** OpenCV, PyTorch, TensorFlow/Keras, YOLO (v5/v8), OCR (EasyOCR), Data Augmentation, Camera Calibration, cv_bridge
- **Tools & Platforms:** Linux, Git, GitHub, Docker, CMake, colcon, ESP-IDF, LaTeX

Professional & Technical Experience

CSIR - National Aerospace Laboratories (NAL)
Research Intern, Project Orion

Bengaluru, India
June 2025 – July 2025

- Architected and validated a multi-sensor EKF-fused SLAM system for an autonomous inspection UGV, **reducing end-of-path trajectory error by over 90%** compared to a VIO-only baseline.
- Engineered a complete perception and control stack on an NVIDIA Jetson Orin Nano, from low-level C++ driver modification to high-level SLAM implementation in ROS 2.
- Solved critical real-world integration challenges, including numerical instability of the IMU and kinematic mismatches, by developing a novel data-driven calibration methodology (SBKC).
- Co-authored the research paper, "AUTOMATING THE VISUAL INSPECTION OF AIRCRAFT STRUCTURES USING 2D AND 3D COMPUTER VISION, MACHINE LEARNING, AND ROBOTICS." published in the *International Conference on Advances in Robotics, Machines and Structures (ARMS 2025)*.

Intel Corporation (Intel Unnati Program)
Industrial Trainee - Quality Control Automation

Kottayam, Kerala, India
May 2025 – July 2025

- Architected a full-stack, ROS 2-based system for AI-driven validation, featuring a PyBullet digital twin, a responsive UI for monitoring, and an SQLite backend for robust data logging.
- Designed and implemented a multi-stage AI pipeline featuring OCR (EasyOCR), QR code decoding, and a custom-trained TensorFlow/Keras model for print quality assessment.
- Achieved **100% precision on "Bad Label" classification** by tuning the ML model's operating threshold, guaranteeing no defective products were passed by the quality check.

Team Equinox (BAJA SAEINDIA)

Vice Captain (2026 Season)

Lead Autonomous Systems Integration Engineer (2025 Season)

Kottayam, Kerala, India

Nov 2025 – Present

Mar 2024 – Oct 2025

- Spearheaded the autonomous systems integration, culminating in an **All India Rank of 7th** at the aBAJA 2025 competition and flawless passes on all technical and autonomous checks on the first attempt.
- Architected the vehicle's distributed, real-time control architecture; integrated a high-level NVIDIA Jetson with multiple low-level Arduino R4 Minima nodes over a robust **CAN bus** network.
- Engineered safety-critical firmware, including PID controllers for the throttle, brake, and steering actuators, ensuring precise vehicle control and adherence to competition rules.

Taiwan-India Smart Cities Project (Saintgits & CCU, Taiwan)

Lead IoT Systems Engineer

Kottayam, India

Aug 2025 – Present

- Coordinated the development and deployment of prototype IoT sensor nodes for the world's first cross-nation federated AI learning platform.
- Engineered the embedded software for ESP32 microcontrollers to reliably publish real-time environmental data via MQTT to the central data server.
- Led a student team in the hands-on building, testing, and debugging of the hardware nodes, ensuring data integrity for the AI backend.

Open Source Contributions

The ROS 2 Community

ROS 2 Contributor

Remote

May 2024 – Present

- Ranked **3rd globally** in the ROS 2 Kilted Kaiju "Tutorial Party" by validating and closing **over 51 critical issues**, receiving official acknowledgment from Open Robotics.
- Actively contribute to the `ros2_control` framework, addressing issues and improving the core control architecture for the ROS 2 community.

MPU6050 Library for Aries Board

Creator & Maintainer

Open Source Project

Jan 2025 – Mar 2025

- Engineered and open-sourced a custom MPU6050 sensor library in C++, acknowledged by Vega Processors for a significant contribution to their RISC-V hardware ecosystem.

Selected Technical Projects

- **Autonomous Lane-Keeping Perception System:** Developed a ROS 2 perception node by integrating the Ultrafast Lane Detection (UFLD) model to accurately identify lane boundaries from real-world video feeds.
- **End-to-End Object Detection Pipeline in Simulation:** Architected and implemented a complete ROS 2 perception pipeline in Gazebo, successfully streaming data from a simulated TurtleBot3 camera to a YOLOv5 node and visualizing real-time object detections in RViz2.
- **Self-Balancing Robot with Kalman Filter:** Engineered a complete control system in C++ for a physical self-balancing robot on a RISC-V board, implementing a Kalman filter and PID controller for stable state estimation.

Leadership & Achievements

- **BAJA SAEINDIA National Competition (2025):** Secured **All India Rank 7** as the Lead Autonomous Systems Integration Engineer for Team Equinox.
- **Aspire Leaders Program (2025):** Selected from over 26,000 global applicants for a competitive leadership program developed by the Aspire Institute with Harvard Business School faculty.
- **University Leadership:** Vice Chairman, The Robotics Society Saintgits Chapter; Co-Lead, TinkerHub Saintgits.